A Structural Local Identifiability Obstruction for the 31-Coefficient v18 Spectral Model

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Abstract

We consider the eleven-nonzero-observable sensitivity map of a spectral model with twenty-nine stored scalar coefficients and two additional historically optimized parameters. At every parameter point at which the 31-parameter forward map is differentiable, its Jacobian has eleven rows, so rank–nullity yields an exact lower bound of twenty on the dimension of its first-order null space. We give a full proof, spell out the precise perturbative interpretation, and distinguish the structural theorem from the repository's finite-difference singular-value-decomposition (SVD) result reporting numerical rank eleven. A block-structured 11-by-11 minor provides an independent numerical diagnostic. We also state the stronger local level-set result that follows *conditionally* from an exact full-row-rank certificate and suitable smoothness. The result is an audit of parameter identifiability, not a derivation of particle masses or an independent physical spectral prediction.

1 Model and hypotheses

Let $U \subseteq \mathbb{R}^{31}$ be an open neighborhood of a reference coefficient vector θ_0, and let

$$F : U \longrightarrow \mathbb{R}^{11}, \qquad F(\theta) = \big(f_1(\theta), \ldots, f_{11}(\theta)\big)^{\mathsf{T}} \tag{1}$$

be the spectral forward map for the eleven nonzero observables. The assumption that F is differentiable at θ_0 is explicit: the implementation includes piecewise routing, clipping, and floors, so differentiability should not be asserted at a branch transition. Within a fixed smooth branch, the calculation below applies directly. The rank–nullity conclusion also applies to any chosen linear sensitivity matrix of the indicated shape, independently of how its entries were estimated.

The audited parameter allocation is

Parameter section	Number of scalar parameters
Light sector	9
Middle sector	7
Heavy sector	13
Historically live suppression parameters	2
Total	31

The v18 source specifies twelve targets, including one artificial zero-mass target. The full-coefficient audit excludes that zero target and uses the index set

$$\mathcal{N} = \{1, 2, 3, 12, 19, 53, 73, 91, 95, 96, 123\}, \qquad |\mathcal{N}| = 11. \tag{2}$$

The corresponding eleven outputs are evaluated without refitting the stored coefficients or reassigning particle labels.

2 Exact structural theorem

Theorem 2.1 (Minimum first-order nullity)**.** *Suppose the map F in* (1) *is differentiable at $\theta_0 \in U$. Its sensitivity matrix*

$$J = DF(\theta_0) = \left(\frac{\partial f_i}{\partial \theta_j}(\theta_0) \right)_{\substack{1 \le i \le 11 \\ 1 \le j \le 31}} \in \mathbb{R}^{11 \times 31} \tag{3}$$

satisfies

$$\boxed{\operatorname{rank} J \le 11}, \qquad \boxed{\dim \ker J = 31 - \operatorname{rank} J \ge 20}. \tag{4}$$

In particular, there are at least twenty linearly independent directions in parameter space along which all eleven predictions have vanishing first directional derivative.

Proof. The derivative $DF(\theta_0)$ is a linear map

$$J : \mathbb{R}^{31} \longrightarrow \mathbb{R}^{11}. \tag{5}$$

Because the dimension of its codomain is eleven,

$$\operatorname{rank} J \le \min\{31, 11\} = 11. \tag{6}$$

The rank–nullity theorem gives

$$\dim \ker J + \operatorname{rank} J = \dim \mathbb{R}^{31} = 31, \tag{7}$$

and hence

$$\dim \ker J = 31 - \operatorname{rank} J \ge 31 - 11 = 20. \tag{8}$$

Choose linearly independent vectors $v_1, \dots, v_d$ forming a basis of $\ker J$, where $d \ge 20$. For any $v \in \ker J$, we have $Jv = 0$. Differentiability entails

$$F(\theta_0 + tv) = F(\theta_0) + tJv + o(|t|) = F(\theta_0) + o(|t|) \quad (t \to 0). \tag{9}$$

Thus the first derivative of every component of F vanishes along each v_i, proving the last assertion. □

Remark 2.2 (Meaning of "unidentifiable direction")**.** *Equation* (9) *is a* first-order *statement; it neither says that a particular parameter is independently dispensable nor that $F(\theta_0 + tv) = F(\theta_0)$ holds for nonzero t. The null vectors are generally combinations of the stored and live coefficients. Furthermore, if the physically admissible parameter set has active bounds, not every signed null perturbation need remain feasible. The theorem concerns the ambient linearized map.*

Corollary 2.3 (Exact nullity under exact full row rank)**.** *If, in addition,* $\operatorname{rank} DF(\theta_0) = 11$ *holds* exactly*, then*

$$\boxed{\dim \ker DF(\theta_0) = 20}. \tag{10}$$

Proof. Insert $\operatorname{rank} J = 11$ into rank–nullity. □

3 Reproducing the numerical sensitivity audit

The full-coefficient script loads the unchanged v18 implementation, checks the original predictions against an independent explicit branch-dispatch evaluator, and applies finite differences about the historical parameter vector θ_0. For a coordinate vector e_j, it usually estimates

$$J_{ij} \approx \frac{f_i(\theta_0 + h_j e_j) - f_i(\theta_0 - h_j e_j)}{2h_j}, \qquad h_j = 10^{-5} \max\{1, |\theta_{0,j}|\}. \tag{11}$$

The historically live parameter ϵ_{cutoff} is at its upper bound 5, so that column is computed using a backward difference. Other active bounds, when relevant, also trigger one-sided differences. These finite differences are diagnostics and are not an exact derivative certificate.

Let $y_i > 0$ denote the eleven observed nonzero target masses and set $s_j = \max\{1, |\theta_{0,j}|\}$. The reported scaled matrix is

$$\widetilde{J} = D_y^{-1} J D_s, \qquad D_y = \operatorname{diag}(y_1, \ldots, y_{11}), \quad D_s = \operatorname{diag}(s_1, \ldots, s_{31}). \tag{12}$$

Both diagonal matrices are invertible, so exact linear-algebraic rank is unchanged:

$$\operatorname{rank} \widetilde{J} = \operatorname{rank} J. \tag{13}$$

However, a numerical rank estimate can depend on scaling, finite differences, and the threshold used in the SVD.

Writing the eleven singular values as $\sigma_1 \geq \cdots \geq \sigma_{11} \geq 0$, the script defines

$$r_{\text{num}} = \#\{i : \sigma_i > 10^{-8} \sigma_1\}. \tag{14}$$

The executable audit reports

$$\boxed{r_{\text{num}} = 11}, \qquad \boxed{31 - r_{\text{num}} = 20}. \tag{15}$$

This is *numerical* full row rank. By contrast, the lower bound $\dim \ker J \geq 20$ in Theorem 2.1 is an exact consequence of dimensions, even if the true rank were below eleven.

4 Independent block-minor diagnostic

The piecewise v18 model allows an independent check of the reported row-rank result. Order the output rows as three light, five middle, and three heavy targets. Select the following eleven *stored* parameter columns:

$$\begin{aligned} \text{light:} &\quad m_0,\ \beta,\ \gamma, \\ \text{middle:} &\quad A,\ \phi,\ h,\ c,\ d, \\ \text{heavy:} &\quad \omega,\ \varphi,\ b_H. \end{aligned}$$

Here γ is the cubic-index coefficient in the light formula, A, ϕ, h, c, d are respectively the middle sinusoidal amplitude, phase, second-harmonic coefficient, constant offset, and quadratic coefficient, and ω, φ, b_H are the W/Z-break angle, phase, and Higgs offset.

At the frozen smooth branches the selected minor has the form

$$J_{\text{minor}} = \begin{pmatrix} L & 0 & 0 \\ 0 & M & 0 \\ B & 0 & H \end{pmatrix}, \qquad \det J_{\text{minor}} = \det L \, \det M \, \det H. \tag{16}$$

The unspecified block B represents dependence of heavy predictions on light-sector coefficients, and does not affect the determinant.

4.1 Light sector

In the v18 light formula, away from an active prediction floor, the selected partial derivatives are a nonzero diagonal row rescaling of

$$L_0 = \begin{pmatrix} 1/2 & 0 & 1 \\ 2/3 & 1 & 8 \\ 3/4 & 1 & 27 \end{pmatrix}. \tag{17}$$

Direct expansion yields

$$\det L_0 = \frac{1}{2}(27-8) + \left(\frac{2}{3} - \frac{3}{4}\right) = \boxed{\frac{113}{12} \neq 0}. \tag{18}$$

The multipliers are the positive logistic suppression factors and the nonzero special $n = 3$ neutrino factor, so $\det L \neq 0$ on the stated unclipped branch.

4.2 Middle sector

The middle mass formula is

$$f(n) = A\sin(sn+\phi) + h\sin(tn) + c + dn^2, \tag{19}$$

up to its inactive positive-mass floor. For $n \in \{12, 19, 53, 73, 91\}$, the five selected derivative columns are

$$M_n = \left(\sin(sn+\phi),\ A\cos(sn+\phi),\ \sin(tn),\ 1,\ n^2\right). \tag{20}$$

Using the stored decimal coefficient values

$$(A, s, t, \phi) = (2241.1, 5.0632, 9.4062, 1.6469), \tag{21}$$

ordinary floating-point evaluation gives

$$\det M \approx 2.1220606421 \times 10^7. \tag{22}$$

This is a numerical nonsingularity witness, not a rigorous interval certificate of the transcendental determinant.

4.3 Heavy sector

The v18 source adds a W/Z correction $S\sin(\omega n + \varphi)$ for $n = 95, 96$, and a separate Higgs offset at $n = 123$. On the corresponding unclipped branches,

$$H = \begin{pmatrix} 95S\cos(95\omega+\varphi) & S\cos(95\omega+\varphi) & 0 \\ 96S\cos(96\omega+\varphi) & S\cos(96\omega+\varphi) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{23}$$

Therefore

$$\det H = -S^2\cos(95\omega+\varphi)\cos(96\omega+\varphi). \tag{24}$$

The source values $S = 8624$, $\omega = -1.264$, and $\varphi = 4.685$ give

$$\det H \approx -4.5126633137 \times 10^7. \tag{25}$$

The two nonzero numerical determinants in (22) and (24), together with the exact light determinant (18), support the numerical full-row-rank result. A formal exact-rank certification can be obtained by replacing these floating-point evaluations with verified interval enclosures that exclude zero, while also verifying the relevant branches remain smooth. This certification is not claimed here.

5 What exact full row rank would additionally prove

Proposition 5.1 (Conditional nonlinear local level set)**.** *Assume F is continuously differentiable on an open neighborhood of θ_0, and assume the* exact *rank condition* $\operatorname{rank} DF(\theta_0) = 11$*. Then, in a sufficiently small neighborhood of θ_0, the level set*

$$\mathcal{M} = \{\theta : F(\theta) = F(\theta_0)\} \tag{26}$$

is a C^1 submanifold of dimension twenty.

Proof. Exact row rank eleven means that $DF(\theta_0)$ is surjective. The submersion (or implicit-function) theorem then provides local coordinates in which F is the projection from $\mathbb{R}^{31}$ onto the first eleven coordinates, up to smooth changes of coordinates. The fiber over $F(\theta_0)$ is locally described by fixing those eleven coordinates, leaving $31 - 11 = 20$ free coordinates. □

Remark 5.2 (Active bounds)**.** *The historical live parameter ϵ_{cutoff} sits at an optimizer upper bound. Proposition 5.1 is therefore stated for a smooth extension to an* open *ambient domain; it does not automatically imply that every direction in the full twenty-dimensional fiber respects the fitted parameter constraints. Since the proposed numerically nonsingular minor in Section 4 uses eleven* stored *parameters, certifying that minor would also permit a restricted implicit-function analysis with the two live parameters fixed. In that case the remaining twenty-nine stored parameters would support a local $29 - 11 = 18$-dimensional equal-prediction level set, without moving either live parameter across its bound.*

6 Scope and scientific conclusion

The theorem makes no claim that exactly twenty individually named coefficients are redundant, that the measured masses identify the remaining eleven coefficients, or that a fitted spectral formula follows from a specified eigenvalue operator. It does not determine measurement uncertainties or Bayesian posterior widths. In particular, it does not supply an independent mapping from eigenmodes to particle assignments, a confinement operator with its domain and boundary conditions, or a physical normalization converting eigenvalues into measured mass units.

The logical status of the main statements is:

Statement	Status
$\operatorname{rank} J \leq 11$ for an 11-by-31 Jacobian	Exact theorem
$\dim \ker J \geq 20$	Exact theorem
At least twenty zero first-order sensitivities	Exact when differentiable
The frozen audit's reported numerical rank eleven	Numerical diagnostic
Exactly twenty-dimensional smooth local fibers	Conditional on exact rank and regularity
Independent spectral prediction of particle masses	Not established

Conclusion. For the unrestricted 31-coefficient parameterization and eleven nonzero observables, the first-order structural obstruction is

$$\boxed{J \in \mathbb{R}^{11\times 31} \quad \Longrightarrow \quad \operatorname{rank} J \leq 11 \quad \Longrightarrow \quad \dim \ker J \geq 20.} \tag{27}$$

The unconditional conclusion is a local sensitivity obstruction, not an independently validated physical spectrum. The corresponding v18 research classification remains `EMPIRICAL_FIT_ONLY`.

Sources and reproducibility

1. v18 audit and scientific-status discussion:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/SM_V18_SPECTRAL_BRIDGE_AUDIT.md`
2. Full-coefficient finite-difference/SVD audit:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/SM_Spectrum/audit_v18_identifiability.py`
3. Original v18 source defining the coefficient blocks and predictions:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/SM_Spectrum/_terms_perfection_v18.py`
4. Foundational Stage 2 spectral-bridge obstruction report:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/FOUNDATIONAL_STAGE2_2026-10-14.md`

The repository's reproduction command is

```
python SM_Spectrum/audit_v18_identifiability.py \
  --output audit/v18_identifiability.json
```

It records coefficient and CSV file hashes, package versions, the 11-by-31 scaled sensitivity, its SVD, and the declared numerical-rank threshold. This document reports the repository's rank result; it does not claim a new executed holdout test.